Part 1 — Problem Framing

Problem C: Competitive Intelligence | upGrad

1. Problem Definition

Problem: In India's competitive online higher-education market, upGrad needs a fast, reliable view of what competitors are doing across pricing, programs, positioning, and marketing. Information is fragmented across websites, program brochures, and advertising platforms, making consistent comparisons time-consuming and difficult to maintain.

Who faces this problem? Strategy, product/category, and marketing teams responsible for tracking the competitive landscape and supporting business decisions.

Current approach (assumption): Without visibility into upGrad's internal processes, we assume competitive research involves manually reviewing competitor websites, program brochures, and advertising libraries, then consolidating findings into reports. This process can be time-consuming and vulnerable to outdated, inconsistent, or incomplete information.

Initial scope: Analyze Great Learning, Simplilearn, and Emeritus against upGrad across two Indian program categories: Data Science & Analytics and Executive / General Management.

Desired output: A standardized, evidence-backed competitive intelligence brief comparing competitors across pricing, programs, positioning, and marketing, with relevant differences highlighted, traceable sources, and clear verification status.

2. Success Metrics

Primary Metric — Time to produce a verified competitor brief

Target: ≥70% reduction in total research and reporting time compared with a manual baseline, using identical programs, information fields, and verification standards.

Guardrail 1 — Factual Accuracy

Target: ≥95% field-level accuracy across a manually verified evaluation dataset, covering pricing, duration, credentials, and other factual program attributes.

Guardrail 2 — Evidence Traceability

Target: 100% of material factual claims must include a traceable source and verification status. Conflicting, missing, or unverified information must be explicitly flagged.

All thresholds are proposed pilot targets, not observed results.

3. Why AI?

Competitor information is largely unstructured and inconsistently presented across different sources. AI is suitable for:

- Extraction: Converting program pages, brochures, and marketing content into structured, comparable information.
- Analysis: Identifying differences in programs, pricing, target audiences, positioning, and marketing messages.
- Synthesis: Producing a concise, readable competitor intelligence brief from multiple sources.

Where AI alone is insufficient: AI can hallucinate information, misinterpret pricing, or rely on outdated content. Source tracking, numerical validation, and change detection should use deterministic methods wherever possible. Ambiguous or conflicting information should be flagged for human verification rather than treated as fact.

Core hypothesis: A human-in-the-loop AI workflow can substantially reduce the time required to produce reliable competitive intelligence without compromising factual accuracy or evidence traceability.

Part 2 — AI-Powered Competitive Research

Company: upGrad
Competitors: Great Learning, Simplilearn, Emeritus
Scope: Data Science & Analytics and Executive / General Management (8 programs)

Five Key Insights

1. Data Science Programs Share Core Topics but Differ in Curriculum Emphasis

All four providers cover Python, statistics, analytics, and machine learning. However, upGrad and Great Learning offer broader data science curricula, while Simplilearn emphasizes analytics tools such as Power BI.

Why it matters: Similar program categories do not necessarily offer equivalent learning experiences.

Sources: D1, D2, D3, D4
Verification: Published curricula reviewed; actual instructional depth not independently verified.

2. Management Programs Differ in Target Audience, Duration, and Learning Format

All four providers cover finance, marketing, strategy, and leadership. However, program durations range from 6 to 12 months. upGrad primarily targets early-career professionals, while competitors serve broader experience levels.

Why it matters: Target audience, duration, and delivery format influence how directly competing programs can be compared.

Sources: M1, M2, M3, M4
Verification: Advertised durations and eligibility requirements reviewed; some delivery details are cohort-specific.

3. Inconsistent Fee Disclosures Make Direct Pricing Comparisons Difficult

For Data Science programs, upGrad advertises ₹1.80 lakh, Simplilearn ₹1.35 lakh (including taxes), and Emeritus ₹1.32 lakh + GST. Great Learning displays conflicting fees of ₹1.95 lakh + GST and ₹2.10 lakh + GST.

Why it matters: Different tax treatments, conflicting prices, and cohort-specific offers prevent reliable like-for-like pricing comparisons.

Sources: D1, D2, D3, D4
Verification: Advertised fees reviewed; final payable amounts and current-cohort applicability remain unconfirmed.

4. Institutional Partnerships Offer Different Credentials and Qualifications

All four providers emphasize partnerships or credentials associated with institutions such as IITs, IIMs, UT Austin, and Great Lakes. However, certificate types, issuing institutions, and alumni benefits differ. The selected management programs award certificates, not MBA degrees.

Why it matters: Institutional associations do not automatically imply equivalent qualifications, recognition, or learner benefits.

Sources: D1, D2, M1, M2, M3, M4
Verification: Published credential information reviewed; some certificate wording and alumni benefits require further confirmation.

5. Competitors Emphasize Different Value Propositions in Their Marketing

Official program messaging indicates different areas of emphasis:

- **upGrad:** Career advancement and university-linked credentials.
- **Great Learning:** Mentorship and career support.
- **Simplilearn:** Practical, job-relevant technical skills.
- **Emeritus:** Executive education and institutional credibility.

Why it matters: Competitors promote overlapping benefits but emphasize different aspects of their offerings to attract learners.

Sources: D1, D2, D3, M4
Verification: Based on published marketing claims and qualitative interpretation. Advertising performance and active ad counts could not be verified.

What the AI got wrong or could not verify

1. Missing source URLs

AI initially provided citation labels instead of direct URLs, making findings difficult to verify.

Correction: Requested an evidence audit and created a source register containing official URLs.

2. Historical pricing presented without sufficient context

AI-assisted research surfaced Simplilearn management pricing from an older IIM Indore cohort, creating a risk of treating historical fees as current.

Correction: Identified the historical cohort and excluded those figures from current-price comparisons.

3. Insufficient marketing evidence

AI could not reliably verify active advertisement counts. Some LinkedIn examples also lacked individual post URLs and confirmed publication dates.

Correction: Excluded unsupported advertising counts and downgraded unverifiable campaign references.

Additional observation: Great Learning's conflicting fees and upGrad's inconsistent program durations originate from their websites, not necessarily AI errors. These discrepancies were preserved and flagged rather than arbitrarily resolved.

Source Register

Data Science & Analytics

D1 — upGrad
https://www.upgrad.com/data-science-pgc-iiitb/

D2 — Great Learning
https://utaustin.mygreatlearning.com/pg-program-data-science-and-business-analytics-course

D3 — Simplilearn
https://www.simplilearn.com/iitk-professional-certificate-course-data-analytics

D4 — Emeritus
https://programs.eictiitg.com/iitg-dsba-tech-certificate-programme

Executive / General Management

M1 — upGrad
https://www.upgrad.com/iim-bangalore-general-management-certificate-for-young-leaders/

M2 — Great Learning
https://www.mygreatlearning.com/pg-program-management-executive

M3 — Simplilearn
https://iimi.simplilearn.com/executive-general-management-program

M4 — Emeritus
https://iimk.emeritus.org/iimk-general-management-programme-for-business-excellence

Additional Source — IIM Indore (Historical Cohort)
https://iimidr.ac.in/programmes/executive-programmes/long-duration-online-programmes-asynchronous/executive-certificate-programme-in-general-management-asynchronous-batch-1/

Research Date: 3 November 2026

Methodology: AI-assisted public-source research, followed by evidence audits and selected source verification. Findings reflect publicly advertised information, not independently confirmed admission offers or educational outcomes.

Part 3 — Working Prototype and Evaluation

Prototype: upGrad Radar
Live application: https://upgrad-radar.vercel.app/
Evaluation date: 4 November 2026

1. What We Built

upGrad Radar is an AI-powered competitive research tool that automates the collection, extraction, validation, and comparison of publicly available competitor information.

Users select a program category, and the tool automatically researches four configured official program pages: upGrad, Great Learning, Simplilearn, and Emeritus.

Core workflow: Live webpage retrieval → Gemini-based structured extraction → source and evidence validation → discrepancy detection → comparative research brief.

Outputs: Side-by-side comparisons of pricing, programs, positioning, and on-page marketing; evidence quotations; verification statuses; and downloadable Markdown reports.

The deployed prototype uses Next.js, TypeScript, direct HTML retrieval, and Gemini-based extraction. Firecrawl is an optional integration but was not used in the evaluated production runs.

2. Evaluation Method and Results

Both program categories were executed against live official webpages. Outputs were compared with our Part 2 reference research and selected original source pages.

Test	Observed result	Status
Live research	Both categories completed; eight program pages retrieved across two runs	Pass
India-specific pricing	Earlier US/USD mismatches corrected; Indian prices recovered	Pass
Conflict detection	Flagged Great Learning's ₹1.95L/₹2.10L + GST fees and upGrad's 6/6–9 month durations	Pass

Historical information	Emeritus's 11 October 2026 orientation marked historical; uncertain cohort information flagged	Pass
Evidence and export	Reports contain source URLs, quotes, timestamps, statuses, and export successfully	Pass, with evidence-quality limitations
Automated reliability tests	Coding agent reported 34 passing automated tests and 61/61 scripted production acceptance checks	Reported pass
Targeted factual audit	12/12 selected material facts and discrepancy flags matched published source information	Pass on selected sample

The Data Science report completed in approximately **24 seconds** and the Management report in **21 seconds**, excluding human verification.

3. Issues Discovered and Corrected

Initial failure: Some official URLs redirected to international pages. This caused USD prices to appear in India-focused comparisons and prevented detection of known Indian pricing discrepancies.

Correction: Added market/program matching and quarantined incompatible regional offers. Subsequent runs recovered Indian prices and correctly flagged conflicting information.

Further improvements addressed missing extraction fields, historical pricing, and false conflicts between complementary credentials or different program variants.

4. Limitations and Conclusion

- A source-backed claim means it appears on an official webpage, not that it has been independently verified as true.
- The 12/12 checks were deliberately selected, not a representative accuracy sample. The ≥95% overall accuracy guardrail remains unproven.
- The ≥70% time-saving target has not been measured against a comparable manual workflow including verification.
- Some tax classifications, field mappings, and supporting evidence remain imperfect. Live outage/recovery testing was not independently completed.
- Marketing analysis covers official landing-page messaging, not verified advertising activity or performance.

Evaluation conclusion: The prototype demonstrates a repeatable, functional AI-assisted research workflow. It is suitable for a controlled, human-reviewed pilot, but not yet for unattended business use.

Part 4 — Recommendation and Next Opportunity

1. Recommendation: ITERATE

Decision: Proceed with a 30-day human-in-the-loop pilot, but do not deploy upGrad Radar as an independently trusted research system yet.

Rationale: The prototype successfully automates retrieval, AI extraction, evidence tracking, and competitor comparisons across eight programs. Initial issues involving international pricing, missing fields, and incorrect conflict detection were identified and substantially corrected.

The updated evaluation reports 34 passing automated tests, 61/61 scripted production acceptance checks, and 12/12 selected factual source checks. However, these results do not establish overall factual accuracy or verified time savings. Minor classification issues and source inconsistencies remain.

Leadership decision: The approach is promising enough to pilot, but wider adoption should depend on measurable reliability and productivity improvements.

2. Proposed 30-Day Pilot

Participants: Two competitive research analysts, one category/strategy manager, and one marketing stakeholder.

Scope: Great Learning, Simplilearn, and Emeritus benchmarked against upGrad across Data Science & Analytics and Executive / General Management.

Period	Plan
Week 1	Establish the manual research baseline, create a representative evaluation dataset, and resolve critical data-quality issues.
Week 2	Run AI-assisted research with analyst verification; record time, accuracy, missing fields, and corrections.
Week 3	Repeat research to evaluate consistency, source changes, and stakeholder usefulness.

Week 4	Compare automated and manual workflows; assess metrics and decide whether to expand, iterate, or stop.

Success criteria — consistent with Part 1

- **Primary metric:** ≥70% reduction in total time to produce a verified competitor brief, including human review.
- **Guardrail 1:** ≥95% field-level factual accuracy against a manually verified evaluation sample.
- **Guardrail 2:** 100% of material factual claims include traceable sources and verification statuses.

Key risks and mitigations: Regional redirects and incorrect program matches require market/identity checks; outdated prices and cohorts require review flags; extraction errors require source-linked evidence and human verification; retrieval/API failures require transparent error handling.

Scale decision: Expand only if the metrics are achieved and no critical accuracy failures remain.

3. Next Big Opportunity — AI Program Opportunity Radar

Concept: Extend competitor intelligence into evidence-backed identification of potential new education programs by combining competitor offerings with verified hiring trends and skills-demand signals.

Go-to-market sketch

- **Target users:** Initially upGrad's category strategy and program development teams; later, other education providers.
- **Value proposition:** Reduce the time needed to identify and evaluate promising course opportunities.
- **Distribution:** Begin with one internal category pilot, demonstrate useful opportunity briefs, then expand to additional teams and test external B2B adoption.
- **Pricing:** Internal pilot first; subsequently test monthly B2B subscriptions based on tracked categories and research volume.
- **Measurement:** Time to opportunity brief, proportion of opportunities accepted for further investigation, and program proposals informed by the research.

Conclusion: Competitive research automation is technically feasible. The immediate priority is proving trusted, repeatable business value before scaling into broader market-opportunity intelligence.

AI Process Logs

Problem selected: Competitive Intelligence (Problem C)
Project: upGrad Radar
AI tools used: ChatGPT, Codex, Gemini 3.5 Flash Lite

1. How I Used AI

I used AI throughout the assignment for research, structuring the problem, prototype development, and evaluation. My approach was iterative: define the task, review AI outputs, identify gaps, refine instructions, and validate important findings against original sources.

Stage	AI's Role	My Decisions and Validation
Part 2: Research	Researched competitors, programs, pricing, credentials, and positioning using public sources.	Selected Great Learning, Simplilearn, and Emeritus; narrowed the analysis to two categories and eight programs; reviewed evidence and finalized five insights.
Part 3: Prototype Planning	Helped define the research workflow and technical specification.	Chose a hybrid approach: predefined official programs for reliable research, with optional discovery for additional competitors.
Part 3: Development	AI coding assistant implemented the web application, retrieval, structured extraction, validation, and Markdown export.	Prioritized live research and evidence traceability over additional features.
Part 3: Evaluation	Assisted with test design, output review, debugging, and automated reliability testing.	Compared generated briefs against Part 2 research, identified reliability issues, and prioritized corrections.

Part 4: Recommendation	Helped structure the pilot and adjacent opportunity.	Chose ITERATE rather than claiming production readiness without sufficient evidence.

2. Where AI Was Wrong and How I Corrected It

Example 1 — International pricing in Indian competitor research

The initial prototype retrieved international versions of some official program pages, producing USD prices instead of Indian fees. This created misleading comparisons and prevented detection of known pricing discrepancies.

Correction: I identified the mismatch by comparing generated briefs against our Part 2 reference research. The coding agent was instructed to validate the intended market and program, flag incompatible regional results, and preserve conflicting Indian prices.

Subsequent runs recovered INR fees and detected the known Great Learning pricing and upGrad duration discrepancies.

Example 2 — Incomplete or unreliable research evidence

During Part 2, AI initially returned incomplete source references, surfaced historical cohort information without sufficient context, and could not substantiate advertising metrics.

Correction: I requested a source audit, excluded historical pricing from current comparisons, and removed unsupported advertising claims.

These corrections reinforced the requirement that unsupported information must be flagged rather than guessed.

3. Evaluation and Learnings

The final prototype successfully executed research across both categories and generated source-linked comparative briefs.

The coding agent reported **34 passing automated tests and 61/61 scripted production acceptance checks**. A separate targeted audit matched **12/12 selected factual checks** against published source information.

These results demonstrate working functionality but do not independently establish the proposed ≥95% overall accuracy or ≥70% time-saving targets. Both require further measurement during the proposed pilot.

Key learning: AI is effective at accelerating unstructured research and extraction, but reliable competitive intelligence also requires source verification, contextual checks, conflict detection, and human judgement.

4. AI Conversation and Development Logs

The following links document the actual prompts, AI responses, iterations, corrections, and implementation process.

Research :
[Insert ChatGPT conversation link]

Prototype planning and product specification:
[Insert ChatGPT conversation link]

Coding, debugging, and evaluation:
[Insert coding-agent conversation link or exported session log]

Live prototype: https://upgrad-radar.vercel.app/